

1. Read the Data

```
#include <iostream>
#include <fstream>
#include <string>
using namespace std;
```

```
struct Person
{
    string ssn;
    int day;
    int month;
    int year;
    string first;
    string last;
    double balance;
};
```

```
int main()
{
    Person people[1000];
    int count = 0;

    ifstream fin;

    fin.open("/home/118/peoplefile1.txt");

    if (fin.fail() == true)
    {
        cout << "File can't open :(" << endl;
    }

    while (fin >> people[count].ssn
           >> people[count].day
           >> people[count].month
           >> people[count].year
           >> people[count].first
           >> people[count].last
           >> people[count].balance)
    {
        count = count + 1;
    }
}
```

```

    }

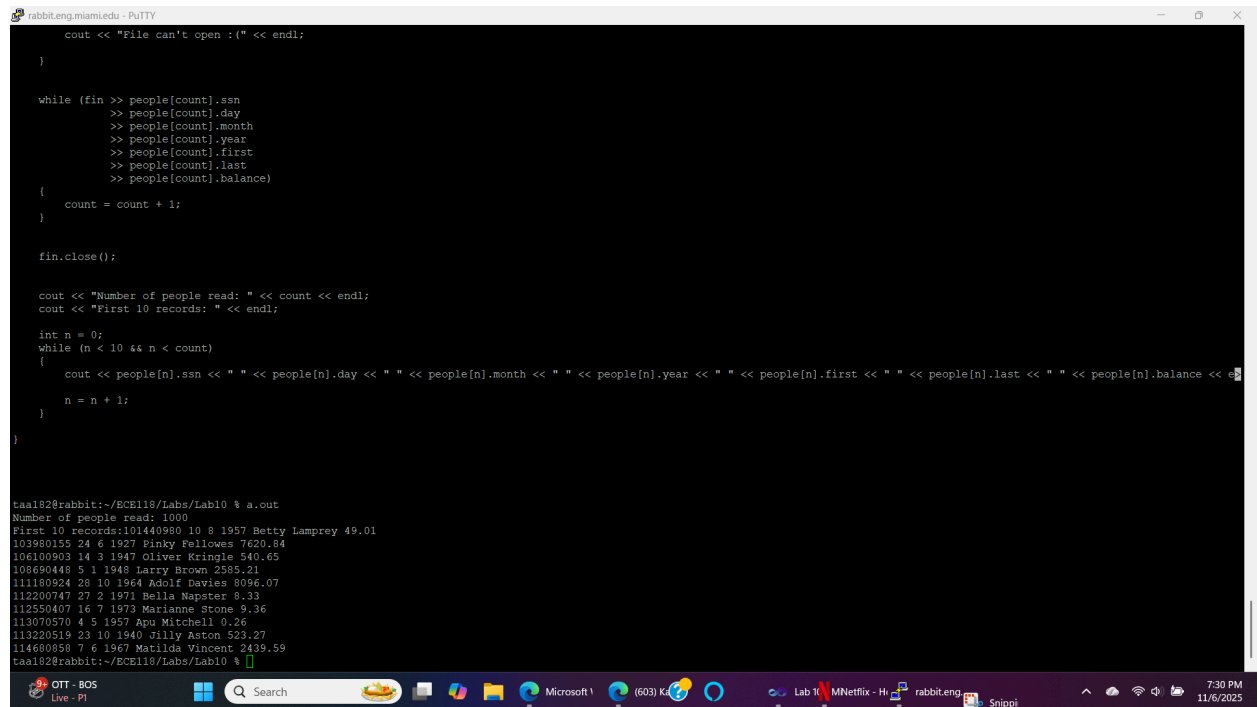
    fin.close();

    cout << "Number of people read: " << count << endl;
    cout << "First 10 records:";

    int n = 0;
    while (n < 10 && n < count)
    {
        cout << people[n].ssn << " " << people[n].day << " " << people[n].month << " " <<
people[n].year << " " << people[n].first << " " << people[n].last << " " << people[n].balance <<
endl;

        n = n + 1;
    }
}

```



```

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cout << "File can't open :(" << endl;
}

while (fin >> people[count].ssn
    >> people[count].day
    >> people[count].month
    >> people[count].year
    >> people[count].first
    >> people[count].last
    >> people[count].balance)
{
    count = count + 1;
}

fin.close();

cout << "Number of people read: " << count << endl;
cout << "First 10 records: " << endl;

int n = 0;
while (n < 10 && n < count)
{
    cout << people[n].ssn << " " << people[n].day << " " << people[n].month << " " << people[n].year << " " << people[n].first << " " << people[n].last << " " << people[n].balance << endl;
    n = n + 1;
}
}

taai82@rabbit:~/ECE110/Labs/Lab10 % a.out
Number of people read: 1000
First 10 records:101440980 10 8 1957 Betty Lamprey 49.01
103980155 24 6 1927 Pinky Fellows 7620.84
106100903 14 3 1947 Oliver Kringle 540.65
108690448 5 1 1948 Larry Brown 2585.21
111180924 28 10 1964 Adolf Davies 8096.07
112200747 27 2 1971 Bella Napster 8.33
112550407 16 7 1973 Marianne Stone 9.36
113070570 4 5 1957 Apu Mitchell 0.26
113220519 23 10 1940 Jilly Aston 523.27
114680858 7 6 1967 Matilda Vincent 2439.59
taai82@rabbit:~/ECE110/Labs/Lab10 %

```

2. Basic search

```
#include <iostream>
#include <fstream>
#include <string>
using namespace std;
```

```
struct Person
{
    string ssn;
    int day;
    int month;
    int year;
    string first;
    string last;
    double balance;
};
```

```
void search(Person people[], int count)
```

```
{
    string name;
    cout << "Enter name: ";
    cin >> name;

    int n = 0;
    int results = 0;

    while (n < count)
    {
        if (people[n].first == name || people[n].last == name)
        {
            cout << people[n].ssn << " " << people[n].day << " " << people[n].month << " " <<
people[n].year << " " << people[n].first << " " << people[n].last << " " << people[n].balance <<
endl;
            results = results + 1;
        }

        n = n + 1;
    }
}
```

```

        if (results == 0)
        {
            cout << "No matches found for " << name << endl;
        }
    }
}

```

```

int main()
{
    Person people[1000];
    int count = 0;

    ifstream fin;

    fin.open("/home/118/peoplefile1.txt");

    if (fin.fail() == true)
    {
        cout << "Can't open file :(" << endl;
    }

    while (fin >> people[count].ssn
           >> people[count].day
           >> people[count].month
           >> people[count].year
           >> people[count].first
           >> people[count].last
           >> people[count].balance)
    {
        count = count + 1;
    }

    fin.close();

    search(people, count);
}

```

```
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    cout << "No matches found for " << name << endl;
}

int main()
{
    Person people[1000];
    int count = 0;

    ifstream fin;
    fin.open("/home/118/peoplefile1.txt");
    if (fin.fail() == true)
    {
        cout << "Can't open file :(" << endl;
    }

    while (fin >> people[count].ssn
           >> people[count].day
           >> people[count].month
           >> people[count].year
           >> people[count].first
           >> people[count].last
           >> people[count].balance)
    {
        count = count + 1;
    }

    fin.close();
    search(people, count);
}

taal82@rabbit:~/ECE118/Labs/Lab10 % CC Lab10.cpp
taal82@rabbit:~/ECE118/Labs/Lab10 % a.out
Enter name: John
672620362 23 8 1980 John Baker 2.57
705670337 25 4 1966 John Newman 105.37
759940467 18 7 1941 John Carman 2507.21
924080608 1 7 1924 John Nelson 90.07
taal82@rabbit:~/ECE118/Labs/Lab10 %
```

Windows taskbar at the bottom shows the time as 9:08 PM on 11/6/2025. Open applications include Downloads, Lab 10 - MS, rabbiteng.n, Microsoft W, and several web browsers like Google Chrome, Microsoft Edge, and Firefox.

3. Find the oldest

```
#include <iostream>
#include <fstream>
#include <string>
using namespace std;
```

```
struct Person
{
    string ssn;
    int day;
    int month;
    int year;
    string first;
    string last;
    double balance;
};
```

```
void oldest_person(Person people[], int count)
{
    int oldest = 0;
    int n = 1;

    while (n < count)
    {
        if (people[n].year < people[oldest].year)
        {
            oldest = n;
        }
        else if (people[n].year == people[oldest].year && people[n].month < people[oldest].month)
        {
            oldest = n;
        }
        else if (people[n].year == people[oldest].year && people[n].month == people[oldest].month
&& people[n].day < people[oldest].day)
        {
            oldest = n;
        }

        n = n + 1;
    }
}
```

```

        cout << "Oldest person: " << endl;
        cout << people[oldest].ssn << " " << people[oldest].day << " " << people[oldest].month << " "
<< people[oldest].year << " " << people[oldest].first << " " << people[oldest].last << " " <<
people[n].balance << endl;
    }

```

```

int main()
{
    Person people[1000];
    int count = 0;

    ifstream fin;

    fin.open("/home/118/peoplefile1.txt");

    if (fin.fail() == true)
    {
        cout << "Can't open file :(" << endl;
    }

    while (fin >> people[count].ssn
           >> people[count].day
           >> people[count].month
           >> people[count].year
           >> people[count].first
           >> people[count].last
           >> people[count].balance)
    {
        count = count + 1;
    }

    fin.close();

    oldest_person(people, count);
}

```

```
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};

void oldest_person(Person people[], int count)
{
    int oldest = 0;
    int n = 1;

    while (n < count)
    {
        if (people[n].year < people[oldest].year)
        {
            oldest = n;
        }
        else if (people[n].year == people[oldest].year && people[n].month < people[oldest].month)
        {
            oldest = n;
        }
        else if (people[n].year == people[oldest].year && people[n].month == people[oldest].month && people[n].day < people[oldest].day)
        {
            oldest = n;
        }

        n = n + 1;
    }

    cout << "Oldest person: " << endl;
    cout << people[oldest].ssn << " " << people[oldest].day << " " << people[oldest].month << " " << people[oldest].year << " " << people[oldest].first << " " << people[oldest].last << " " << endl;
}

int main()
{
    Person people[1000];
    int count = 0;

    ifstream fin;

    fin.open("/home/118/peoplefile1.txt");

    if (fin.fail() == true)

taa182@rabbit:~/ECE118/Labs/Lab10 % CC Lab10.cpp
taa182@rabbit:~/ECE118/Labs/Lab10 % a.out
Oldest person:
755236789 2 5 1896 Cornelius VanDerHefferlump 0.89
taa182@rabbit:~/ECE118/Labs/Lab10 %
```


4. Promote the youngest

```
#include <iostream>
#include <fstream>
#include <string>
using namespace std;
```

```
struct Person
{
    string ssn;
    int day;
    int month;
    int year;
    string first;
    string last;
    double balance;
};
```

```
int youngest_person(Person people[], int count)
{
    int youngest = 0;
    int n = 1;

    while (n < count)
    {
        if (people[n].year > people[youngest].year)
        {
            youngest = n;
        }
        else if (people[n].year == people[youngest].year && people[n].month >
people[youngest].month)
        {
            youngest = n;
        }
        else if (people[n].year == people[youngest].year && people[n].month ==
people[youngest].month && people[n].day > people[youngest].day)
        {
            youngest = n;
        }
    }
}
```

```

        n = n + 1;
    }

    cout << "The youngest person has moved to first position (index 0): " << endl;
    cout << people[youngest].ssn << " " << people[youngest].day << " " <<
    people[youngest].month << " " << people[youngest].year << " " << people[youngest].first << " "
    << people[youngest].last << " " << people[youngest].balance << endl;

}

```

```

void swapping(Person &a, Person &b)
{
    Person temp;
    temp = a;
    a = b;
    b = temp;
}

```

```

int main()
{
    Person people[1000];
    int count = 0;

    ifstream fin;

    fin.open("/home/118/peoplefile1.txt");

    if (fin.fail() == true)
    {
        cout << "Can't open file :(" << endl;
    }
}

```

```

while (fin >> people[count].ssn
    >> people[count].day
    >> people[count].month
    >> people[count].year
    >> people[count].first
    >> people[count].last
    >> people[count].balance)
{
    count = count + 1;
}

```

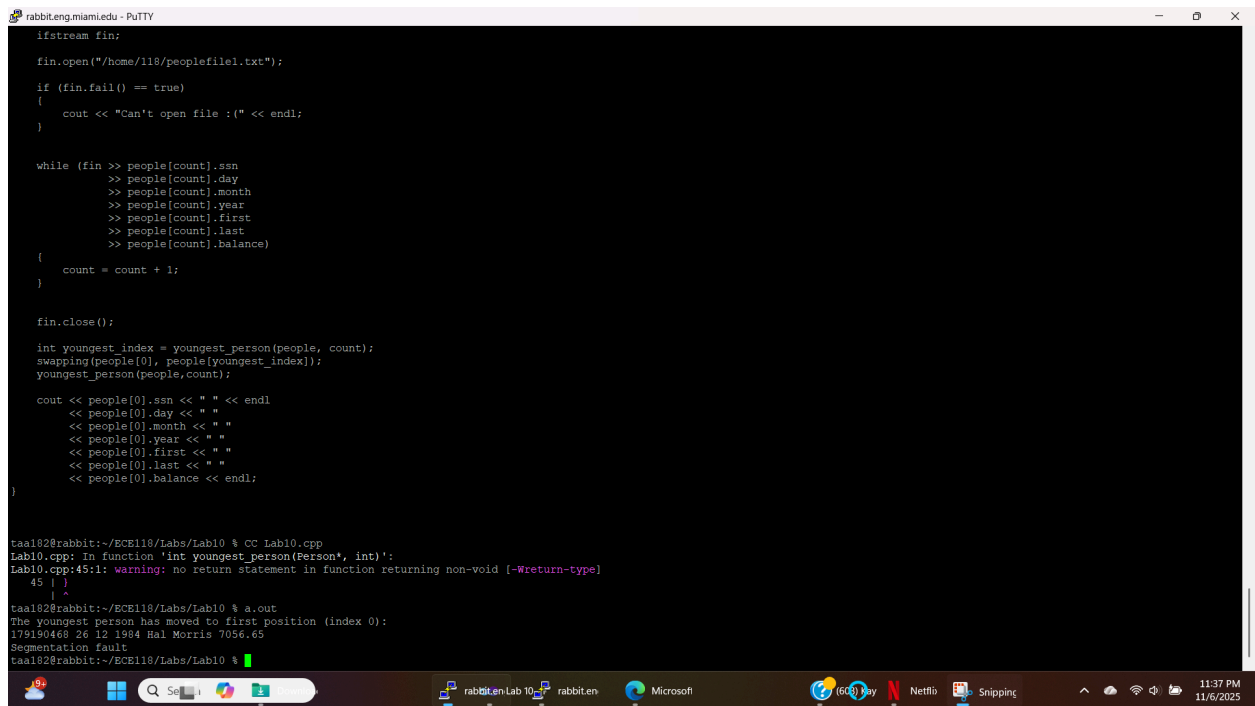
```
}
```

```
fin.close();
```

```
int youngest_index = youngest_person(people, count);  
swapping(people[0], people[youngest_index]);  
youngest_person(people, count);
```

```
cout << people[0].ssn << " " << endl  
    << people[0].day << " "  
    << people[0].month << " "  
    << people[0].year << " "  
    << people[0].first << " "  
    << people[0].last << " "  
    << people[0].balance << endl;
```

```
}
```



```
rabbiteng.miamiedu - PuTTY  
  
ifstream fin;  
  
fin.open("/home/118/peoplefile1.txt");  
  
if (fin.fail() == true)  
{  
    cout << "Can't open file :(" << endl;  
}  
  
while (fin >> people[count].ssn  
    >> people[count].day  
    >> people[count].month  
    >> people[count].year  
    >> people[count].first  
    >> people[count].last  
    >> people[count].balance)  
{  
    count = count + 1;  
}  
  
fin.close();  
  
int youngest_index = youngest_person(people, count);  
swapping(people[0], people[youngest_index]);  
youngest_person(people, count);  
  
cout << people[0].ssn << " " << endl  
    << people[0].day << " "  
    << people[0].month << " "  
    << people[0].year << " "  
    << people[0].first << " "  
    << people[0].last << " "  
    << people[0].balance << endl;  
}  
  
taa182@rabbit:~/ECE118/Labs/Lab10 % CC Lab10.cpp  
Lab10.cpp: In function 'int youngest_person(Person*, int)':  
Lab10.cpp:45:1: warning: no return statement in function returning non-void [-Wreturn-type]  
    45 | }  
      | ^  
taa182@rabbit:~/ECE118/Labs/Lab10 % a.out  
The youngest person has moved to first position (index 0):  
19190469 26 12 1984 Hal Morris 7056.65  
Segmentation fault  
taa182@rabbit:~/ECE118/Labs/Lab10 %
```

5. Now promote the second youngest

```
#include <iostream>
#include <fstream>
#include <string>
using namespace std;
```

```
struct Person
{
    string ssn;
    int day;
    int month;
    int year;
    string first;
    string last;
    double balance;
};
```

```
int age_range(Person people[], int start, int end)
{
    int youngest = start;
    int n = start + 1;

    while (n <= end)
    {
        if (people[n].year > people[youngest].year)
        {
            youngest = n;
        }
        else if (people[n].year == people[youngest].year && people[n].month >
people[youngest].month)
        {
            youngest = n;
        }
        else if (people[n].year == people[youngest].year && people[n].month ==
people[youngest].month && people[n].day > people[youngest].day)
        {
            youngest = n;
        }

        n = n + 1;
    }
}
```

```
    return youngest;
}
```

```
void swapping(Person &a, Person &b)
{
    Person temp;
    temp = a;
    a = b;
    b = temp;
}
```

```
int main()
{
    Person people[1000];
    int count = 0;

    ifstream fin;

    fin.open("/home/118/peoplefile1.txt");

    if (fin.fail() == true)
    {
        cout << "Can't open file :(" << endl;
    }
}
```

```
while (fin >> people[count].ssn
      >> people[count].day
      >> people[count].month
      >> people[count].year
      >> people[count].first
      >> people[count].last
      >> people[count].balance)
{
    count = count + 1;
}
```

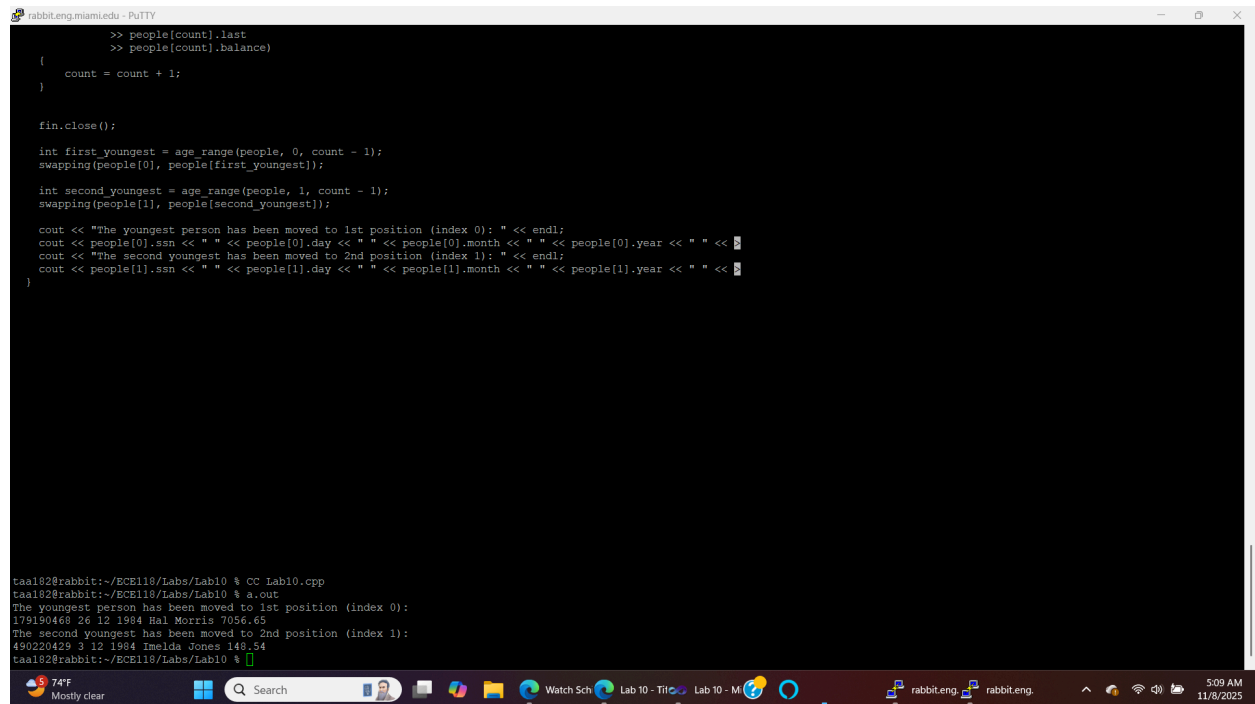
```
fin.close();
```

```
int first_youngest = age_range(people, 0, count - 1);
```

```
swapping(people[0], people[first_youngest]);
```

```
int second_youngest = age_range(people, 1, count - 1);  
swapping(people[1], people[second_youngest]);
```

```
cout << "The youngest person has been moved to 1st position (index 0): " << endl;  
cout << people[0].ssn << " " << people[0].day << " " << people[0].month << " " <<  
people[0].year << " " << people[0].first << " " << people[0].last << " " << people[0].balance <<  
endl;  
cout << "The second youngest has been moved to 2nd position (index 1): " << endl;  
cout << people[1].ssn << " " << people[1].day << " " << people[1].month << " " <<  
people[1].year << " " << people[1].first << " " << people[1].last << " " << people[1].balance <<  
endl;  
  
}
```



The screenshot shows a PuTTY terminal window titled "rabbit.eng.miami.edu - PuTTY". The terminal displays the following C++ code and its output:

```
>> people[count].last  
>> people[count].balance  
{  
    count = count + 1;  
}  
  
fin.close();  
  
int first_youngest = age_range(people, 0, count - 1);  
swapping(people[0], people[first_youngest]);  
  
int second_youngest = age_range(people, 1, count - 1);  
swapping(people[1], people[second_youngest]);  
  
cout << "The youngest person has been moved to 1st position (index 0): " << endl;  
cout << people[0].ssn << " " << people[0].day << " " << people[0].month << " " << people[0].year << " " <<  
cout << "The second youngest has been moved to 2nd position (index 1): " << endl;  
cout << people[1].ssn << " " << people[1].day << " " << people[1].month << " " << people[1].year << " " <<  
}
```

The terminal output shows the execution of the program:

```
taal82@rabbit:~/ECE118/Labs/Lab10 $ CC Lab10.cpp  
taal82@rabbit:~/ECE118/Labs/Lab10 $ a.out  
The youngest person has been moved to 1st position (index 0):  
179190468 26 12 1984 Hal Morris 7056.65  
The second youngest has been moved to 2nd position (index 1):  
490220429 3 12 1984 Imelda Jones 149.54  
taal82@rabbit:~/ECE118/Labs/Lab10 $
```

The terminal window is running on a Windows 10 desktop. The taskbar at the bottom shows the Start button, a search bar, and several open applications including "Watch Sch...", "Lab 10 - Tit...", "Lab 10 - M...", and two instances of "rabbit.eng.". The system tray on the right shows the date and time as "5:09 AM 11/8/2025".

6. More of the same

```
#include <iostream>
#include <fstream>
#include <string>
using namespace std;
```

```
struct Person
{
    string ssn;
    int day;
    int month;
    int year;
    string first;
    string last;
    double balance;
};
```

```
int age_range(Person people[], int start, int end)
{
    int youngest = start;
    int n = start + 1;

    while (n <= end)
    {
        if (people[n].year > people[youngest].year)
        {
            youngest = n;
        }
        else if (people[n].year == people[youngest].year && people[n].month >
people[youngest].month)
        {
            youngest = n;
        }
        else if (people[n].year == people[youngest].year && people[n].month ==
people[youngest].month && people[n].day > people[youngest].day)
        {
            youngest = n;
        }

        n = n + 1;
    }
}
```

```
    return youngest;
}
```

```
void swapping(Person &a, Person &b)
{
    Person temp;
    temp = a;
    a = b;
    b = temp;
}
```

```
int main()
{
    Person people[1000];
    int count = 0;

    ifstream fin;

    fin.open("/home/118/peoplefile1.txt");

    if (fin.fail() == true)
    {
        cout << "Can't open file :(" << endl;
    }
}
```

```
while (fin >> people[count].ssn
        >> people[count].day
        >> people[count].month
        >> people[count].year
        >> people[count].first
        >> people[count].last
        >> people[count].balance)
{
    count = count + 1;
}
```

```
fin.close();
```

```
int first_youngest = age_range(people, 0, count - 1);
```



```

swapping(people[0], people[first_youngest]);

int second_youngest = age_range(people, 1, count - 1);
swapping(people[1], people[second_youngest]);

int third_youngest = age_range(people, 2, count - 1);
swapping(people[2], people[third_youngest]);

cout << "The youngest person has been moved to 1st position (index 0): " << endl;
cout << people[0].ssn << " " << people[0].day << " " << people[0].month << " " <<
people[0].year << " " << people[0].first << " " << people[0].last << " " << people[0].balance <<
endl;
cout << "The second youngest person has been moved to 2nd position (index 1): " << endl;
cout << people[1].ssn << " " << people[1].day << " " << people[1].month << " " <<
people[1].year << " " << people[1].first << " " << people[1].last << " " << people[1].balance <<
endl;
cout << "The third youngest person has been moved to 3rd position (index 2): " << endl;
cout << people[2].ssn << " " << people[2].day << " " << people[2].month << " " <<
people[2].year << " " << people[2].first << " " << people[2].last << " " << people[2].balance <<
endl;>

}

```

```

rabbitt.eng.miami.edu - PuTTY

fin.open("/home/118/peoplefile1.txt");
if (fin.fail() == true)
{
    cout << "Can't open file :(" << endl;
}

while (fin >> people[count].ssn
    >> people[count].day
    >> people[count].month
    >> people[count].year
    >> people[count].first
    >> people[count].last
    >> people[count].balance)
{
    count = count + 1;
}

fin.close();

int first_youngest = age_range(people, 0, count - 1);
swapping(people[0], people[first_youngest]);

int second_youngest = age_range(people, 1, count - 1);
swapping(people[1], people[second_youngest]);

int third_youngest = age_range(people, 2, count - 1);
swapping(people[2], people[third_youngest]);

cout << "The youngest person has been moved to 1st position (index 0): " << endl;
cout << people[0].ssn << " " << people[0].day << " " << people[0].month << " " << people[0].year << " " <<
cout << "The second youngest person has been moved to 2nd position (index 1): " << endl;
cout << people[1].ssn << " " << people[1].day << " " << people[1].month << " " << people[1].year << " " <<
cout << "The third youngest person has been moved to 3rd position (index 2): " << endl;
cout << people[2].ssn << " " << people[2].day << " " << people[2].month << " " << people[2].year << " " <<

taal82@rabbitt:~/ECE118/Labs/Lab10 $ cc Lab10.cpp
taal82@rabbitt:~/ECE118/Labs/Lab10 $ a.out
The youngest person has been moved to 1st position (index 0):
179190468 26 12 1984 Hal Morris 7056.65
The second youngest person has been moved to 2nd position (index 1):
490220429 3 12 1984 Imelda Jones 148.54
The third youngest person has been moved to 3rd position (index 2):
372190403 2 11 1984 Anabel Mosley 6164.29
taal82@rabbitt:~/ECE118/Labs/Lab10 $

```

7. The ultimate demand

```
#include <iostream>
#include <fstream>
#include <string>
using namespace std;
```

```
struct Person
{
    string ssn;
    int day;
    int month;
    int year;
    string first;
    string last;
    double balance;
};
```

```
int age_range(Person people[], int start, int end)
{
    int youngest = start;
    int n = start + 1;

    while (n <= end)
    {
        if (people[n].year > people[youngest].year)
        {
            youngest = n;
        }
        else if (people[n].year == people[youngest].year && people[n].month >
people[youngest].month)
        {
            youngest = n;
        }
        else if (people[n].year == people[youngest].year && people[n].month ==
people[youngest].month && people[n].day > people[youngest].day)
        {
            youngest = n;
        }

        n = n + 1;
    }
}
```

```
    return youngest;
}
```

```
void swapping(Person &a, Person &b)
{
    Person temp;
    temp = a;
    a = b;
    b = temp;
}
```

```
int main()
{
    Person people[1000];
    int count = 0;

    ifstream fin;

    fin.open("/home/118/peoplefile1.txt");

    if (fin.fail() == true)
    {
        cout << "Can't open file :(" << endl;
    }
}
```

```
while (fin >> people[count].ssn
        >> people[count].day
        >> people[count].month
        >> people[count].year
        >> people[count].first
        >> people[count].last
        >> people[count].balance)
{
    count = count + 1;
}
```

```
fin.close();
```

```
int start = 0;
```

```

while (start < count - 1)
{
    int youngest = age_range(people, start, count - 1);
    swapping(people[start], people[youngest]);
    start = start + 1;
}

cout << "Everyone in order from youngest to oldest: " << endl;

int n = 0;
while (n < count)
{
    cout << people[n].ssn << " " << people[n].day << " " << people[n].month << " " <<
people[n].year << " " << people[n].first << " " << people[n].last << " " << people[n].balance << endl;
    n = n + 1;
}
}

```

```

rabbit.eng.miami.edu - PuTTY
143220526 9 11 1922 George Mantovani 8546.79
352600632 8 11 1922 Nellie Reese 136.62
538230470 28 9 1922 Edwina Ellesmere 409.69
540470745 31 7 1922 Katherine Gordon 11.39
545320846 24 7 1922 Fleur Gimble 275.66
832450934 18 6 1922 Cary Newton 505.13
905160869 16 6 1922 Ruby Dimaggio 970.73
903450405 9 5 1922 Joyce Peters 8.07
352190500 12 4 1922 Alan Jameson 89.07
399800089 5 4 1922 Bolton Armstrong 1394.86
833960493 6 3 1922 Brenda Ride 2.65
730430555 18 2 1922 Caroline Oxtable 9.65
451580426 14 2 1922 Pablo Hardy 22.6
123040628 13 2 1922 Elizabeth Watson 32.4
513850964 8 2 1922 Jack Moore 300.68
351560191 29 1 1922 Unicorn Klein 9649.27
894150659 10 1 1922 Frances Krone 6.11
386820868 30 12 1921 Timothy Mack 6733.22
159440684 25 12 1921 Rupert Oxtable 336.61
681630083 11 12 1921 Halle Wine 338.94
184180508 22 11 1921 Rabitha Benson 946.72
991680159 1 11 1921 Pip Ponsonby 91.46
972050440 21 10 1921 Ernest Grimes 0.36
389310604 1 10 1921 Rory Stone 54.3
146190491 26 8 1921 Wally Watson 47.48
922640379 25 5 1921 Wilfred Wallace 42.03
187590396 20 5 1921 Edward Square 2390.93
240460183 1 5 1921 Tachometer Aston 292.07
351900345 4 4 1921 Lillian Cronkite 629.12
380190093 1 4 1921 Incitatus Mantovani 5822.36
704350186 21 2 1921 Torso Alderley 60.54
267140663 16 1 1921 Matilda Patterson 2195.2
217610262 26 12 1920 Hal McAdam 3723.43
815920447 2 12 1920 Hellen Lorton 9.57
634260671 6 11 1920 Georgia Abbot 5.98
238310417 28 8 1920 Yolanda Hoolighan 1773.86
141320100 26 7 1920 Jughead Nippy 1535.62
146680961 2 7 1920 Dick Silverman 6333.57
647100561 28 6 1920 Matthew VanDyke 17.3
483600673 14 6 1920 Ethan Younger 12.58
154100255 14 6 1920 Patrick Lovel 60.78
618170406 25 5 1920 David Freeland 4011.3
332040687 16 5 1920 Trixie Underwood 500.67
544040950 8 4 1920 Nelson Wilde 2.2
671250249 7 2 1920 Matthew Farmer 51.12
149670209 20 1 1920 Yellow Bottomley 8502.8
827376261 20 4 1911 Cerebella Stoot 5.99
765236789 2 5 1896 Cornelius VanDerHefferlump 0.89
taal828rabbit:~/ECE118/Labs/Lab10 %

```

8. Sorting the file

```
#include <iostream>
#include <fstream>
#include <string>
using namespace std;
```

```
struct Person
{
    string ssn;
    int day;
    int month;
    int year;
    string first;
    string last;
    double balance;
};
```

```
int age_range(Person people[], int start, int end)
{
    int youngest = start;
    int n = start + 1;

    while (n <= end)
    {
        if (people[n].year > people[youngest].year)
        {
            youngest = n;
        }
        else if (people[n].year == people[youngest].year && people[n].month >
people[youngest].month)
        {
            youngest = n;
        }
        else if (people[n].year == people[youngest].year && people[n].month ==
people[youngest].month && people[n].day > people[youngest].day)
        {
            youngest = n;
        }

        n = n + 1;
    }
}
```

```
    return youngest;
}
```

```
void swapping(Person &a, Person &b)
{
    Person temp;
    temp = a;
    a = b;
    b = temp;
}
```

```
int main()
{
    Person people[1000];
    int count = 0;

    ifstream fin;

    fin.open("/home/118/peoplefile1.txt");

    if (fin.fail() == true)
    {
        cout << "Can't open file :(" << endl;
    }
}
```

```
while (fin >> people[count].ssn
        >> people[count].day
        >> people[count].month
        >> people[count].year
        >> people[count].first
        >> people[count].last
        >> people[count].balance)
{
    count = count + 1;
}
```

```
fin.close();
```

```
int start = 0;
```

```

while (start < count - 1)
{
    int youngest = age_range(people, start, count - 1);
    swapping(people[start], people[youngest]);
    start = start + 1;
}

ofstream fout;
fout.open("sorted_peoplefile1.txt");

int n = 0;
while (n < count)
{
    cout << people[n].ssn << " " << people[n].day << " " << people[n].month << " " <<
people[n].year << " " << people[n].first << " " << people[n].last << " " << people[n].balance << endl;
    n = n + 1;
}

fout.close();

cout << "Sorted data saved to sorted_peoplefile1.txt" << endl;
}

```

The screenshot shows a Windows desktop environment. A PuTTY window titled "rabbit.eng.miami.edu - PuTTY" is open, displaying a terminal window. Inside the terminal, a nano editor is editing a file named "sorted_peoplefile1.txt". The file contains a list of people's data, sorted by age, with columns for ssn, day, month, year, first, last, and balance. The data is as follows:

ssn	day	month	year	first	last	balance
845320946	24	7	1922	Fleur	Gimble	275.66
832450934	18	6	1922	Cary	Newton	505.13
905160869	16	6	1922	Ruby	Dimaggio	970.73
903450405	9	5	1922	Joyce	Peters	8.07
352190500	12	4	1922	Alan	Jameson	89.07
399800089	5	4	1922	Horton	Armstrong	1394.86
853960493	6	3	1922	Brenda	Ride	2.65
730430555	18	2	1922	Caroline	Oxtable	9.65
451580426	14	2	1922	Pablo	Hardy	22.6
123040628	13	2	1922	Elizabeth	Watson	32.4
513850964	8	2	1922	Jack	Moore	300.68
351560191	29	1	1922	Unicorn	Klein	9649.27
894150659	10	1	1922	Frances	Krone	6.11
386820868	30	12	1921	Tomothy	Mack	6733.22
159440884	25	12	1921	Rupert	Oxtable	336.61
681630083	11	12	1921	Hallie	Wise	3358.84
184180509	22	11	1921	Tabitha	Benson	946.72
991680159	1	11	1921	Pip	Ponsonby	91.46
972050440	21	10	1921	Ernest	Grimes	0.36
389310604	1	10	1921	Rory	Stone	54.3
146190491	26	8	1921	Wally	Watson	47.48
922640579	25	5	1921	Wilfred	Wallace	42.03
187590396	20	5	1921	Edward	Square	2390.93
240460183	1	5	1921	Tachometer	Aston	292.07
351900345	4	4	1921	Lillian	Cronkite	629.12
380190093	1	4	1921	Incitatus	Mantovani	5822.36
704350186	21	2	1921	Torso	Alderley	60.54
267140663	16	1	1921	Matilda	Patterson	2195.2
217610262	26	12	1920	Hal	McAdam	3723.43
815920447	2	12	1920	Hellen	Lorson	9.57
634260671	6	11	1920	Georgia	Abbot	5.98
238310417	28	8	1920	Yolanda	Hoolighan	1773.86
141320100	26	7	1920	Jughead	Nippy	1535.62
146680961	2	7	1920	Dick	Silverman	6333.57
647100561	28	6	1920	Matthew	VanDyke	17.3
483600673	14	6	1920	Ethan	Younger	12.58
154100259	14	6	1920	Patrick	Lovel	60.78
618170406	25	5	1920	David	Freelander	4011.3
332040687	16	5	1920	Trixie	Underwood	500.67
544040950	8	4	1920	Nelson	Wilde	2.2
671250245	7	2	1920	Matthew	Farmer	51.12
149670209	20	1	1920	Yellow	Bottomley	8502.8
927376261	20	4	1911	Cerebella	Stoot	5.99
765236789	2	5	1896	Cornelius	VanDerHefferlump	0.89

The taskbar at the bottom of the screen shows several open applications: Watch Schitt, Microsoft Word, Lab 10 - Micro, and three instances of rabbit.eng.miami.edu. The system clock in the bottom right corner indicates the time is 7:46 AM on 11/8/2025.

9. How fast is it?

/home/118/peoplefile1.txt:

```
#include <iostream>
#include <fstream>
#include <string>
#include <time.h>
#include <sys/resource.h>
using namespace std;

double get_cpu_time()
{ struct rusage ruse;
  getrusage(RUSAGE_SELF, &ruse);
  return ruse.ru_utime.tv_sec+ruse.ru_utime.tv_usec/1000000.0 +
    ruse.ru_stime.tv_sec+ruse.ru_stime.tv_usec/1000000.0; }

struct Person
{
  string ssn;
  int day;
  int month;
  int year;
  string first;
  string last;
  double balance;
};

int age_range(Person people[], int start, int end)
{
  int youngest = start;
  int n = start + 1;

  while (n <= end)
  {
    if (people[n].year > people[youngest].year)
    {
      youngest = n;
    }
    else if (people[n].year == people[youngest].year && people[n].month >
people[youngest].month)
```



```

        {
            youngest = n;
        }
        else if (people[n].year == people[youngest].year && people[n].month ==
people[youngest].month && people[n].day > people[youngest].day)
        {
            youngest = n;
        }

        n = n + 1;
    }

    return youngest;
}

```

```

void swapping(Person &a, Person &b)
{
    Person temp;
    temp = a;
    a = b;
    b = temp;
}

```

```

int main()
{
    Person people[1000000];
    int count = 0;

    ifstream fin;

    fin.open("/home/118/peoplefile1.txt");

    if (fin.fail() == true)
    {
        cout << "Can't open file :(" << endl;
    }

    while (fin >> people[count].ssn
           >> people[count].day
           >> people[count].month

```

```

        >> people[count].year
        >> people[count].first
        >> people[count].last
        >> people[count].balance)
    {
        count = count + 1;
    }

    fin.close();

    double starting_time = get_cpu_time();

    int start = 0;
    while (start < count - 1)
    {
        int youngest = age_range(people, start, count - 1);
        swapping(people[start], people[youngest]);
        start = start + 1;
    }

    double end_time = get_cpu_time();

    cout << "Sorting completed in " << (end_time - starting_time) << " seconds" << endl;
}

```

```

rabbit.eng.miamledu - PuTTY
int count = 0;
ifstream fin;
fin.open("/home/118/peoplefile1.txt");
if (fin.fail() == true)
{
    cout << "Can't open file :(" << endl;
}

while (fin >> people[count].ssn
>> people[count].day
>> people[count].month
>> people[count].year
>> people[count].first
>> people[count].last
>> people[count].balance)
{
    count = count + 1;
}

fin.close();

double starting_time = get_cpu_time();

int start = 0;
while (start < count - 1)
{
    int youngest = age_range(people, start, count - 1);
    swapping(people[start], people[youngest]);
    start = start + 1;
}

double end_time = get_cpu_time();

cout << "Sorting completed in " << (end_time - starting_time) << " seconds" << endl;
}

taai82@rabbit:~/ECE118/Labs/Lab10 % CC Lab10.cpp
taai82@rabbit:~/ECE118/Labs/Lab10 % a.out
Sorting completed in 0.003995 seconds
taai82@rabbit:~/ECE118/Labs/Lab10 %

```

/home/118/peoplefile2.txt:

```
#include <iostream>
#include <fstream>
#include <string>
#include <time.h>
#include <sys/resource.h>
using namespace std;
```

```
double get_cpu_time()
{ struct rusage ruse;
  getrusage(RUSAGE_SELF, &ruse);
  return ruse.ru_utime.tv_sec+ruse.ru_utime.tv_usec/1000000.0 +
    ruse.ru_stime.tv_sec+ruse.ru_stime.tv_usec/1000000.0; }
```

```
struct Person
{
  string ssn;
  int day;
  int month;
  int year;
  string first;
  string last;
  double balance;
};
```

```
int age_range(Person people[], int start, int end)
{
  int youngest = start;
  int n = start + 1;

  while (n <= end)
  {
    if (people[n].year > people[youngest].year)
    {
      youngest = n;
    }
    else if (people[n].year == people[youngest].year && people[n].month >
people[youngest].month)
    {
      youngest = n;
    }
  }
}
```

```

    }
    else if (people[n].year == people[youngest].year && people[n].month ==
people[youngest].month && people[n].day > people[youngest].day)
    {
        youngest = n;

    }

    n = n + 1;
}

return youngest;
}

```

```

void swapping(Person &a, Person &b)
{
    Person temp;
    temp = a;
    a = b;
    b = temp;
}

```

```

int main()
{
    Person people[1000000];
    int count = 0;

    ifstream fin;

    fin.open("/home/118/peoplefile2.txt");

    if (fin.fail() == true)
    {
        cout << "Can't open file :(" << endl;
    }

```

```

    while (fin >> people[count].ssn
        >> people[count].day
        >> people[count].month
        >> people[count].year
        >> people[count].first

```

```

        >> people[count].last
        >> people[count].balance)
    {
        count = count + 1;
    }

    fin.close();

    double starting_time = get_cpu_time();

    int start = 0;
    while (start < count - 1)
    {
        int youngest = age_range(people, start, count - 1);
        swapping(people[start], people[youngest]);
        start = start + 1;
    }

    double end_time = get_cpu_time();

    cout << "Sorting completed in " << (end_time - starting_time) << " seconds" << endl;
}

```

```

rabbiteing.miamiedu - PuTTY
{
    if (people[n].year > people[youngest].year)
    {
        youngest = n;
    }
    else if (people[n].year == people[youngest].year && people[n].month > people[youngest].month)
    {
        youngest = n;
    }
    else if (people[n].year == people[youngest].year && people[n].month == people[youngest].month && people[n].last > people[youngest].last)
    {
        youngest = n;
    }
    n = n + 1;
}
return youngest;
}

void swapping(Person &a, Person &b)
{
    Person temp;
    temp = a;
    a = b;
    b = temp;
}

int main()
{
    Person people[1000000];
    int count = 0;

    ifstream fin;

    fin.open("/home/118/peoplefile2.txt");
    if (fin.fail() == true)
    {
        cout << "Can't open file :(" << endl;
    }

    taal82@rabbitt:~/ECE118/Labs/Lab10 $ CC Lab10.cpp
    taal82@rabbitt:~/ECE118/Labs/Lab10 $ a.out
    Sorting completed in 0.016449 seconds
    taal82@rabbitt:~/ECE118/Labs/Lab10 $

```

/home/118/peoplefile50.txt:

```
#include <iostream>
#include <fstream>
#include <string>
#include <time.h>
#include <sys/resource.h>
using namespace std;
```

```
double get_cpu_time()
{ struct rusage ruse;
  getrusage(RUSAGE_SELF, &ruse);
  return ruse.ru_utime.tv_sec+ruse.ru_utime.tv_usec/1000000.0 +
    ruse.ru_stime.tv_sec+ruse.ru_stime.tv_usec/1000000.0; }
```

```
struct Person
{
    string ssn;
    int day;
    int month;
    int year;
    string first;
    string last;
    double balance;
};
```

```
int age_range(Person people[], int start, int end)
{
    int youngest = start;
    int n = start + 1;

    while (n <= end)
    {
        if (people[n].year > people[youngest].year)
        {
            youngest = n;
        }
        else if (people[n].year == people[youngest].year && people[n].month >
people[youngest].month)
        {
            youngest = n;
        }
    }
}
```

```

    }
    else if (people[n].year == people[youngest].year && people[n].month ==
people[youngest].month && people[n].day > people[youngest].day)
    {
        youngest = n;

    }

    n = n + 1;
}

return youngest;
}

```

```

void swapping(Person &a, Person &b)
{
    Person temp;
    temp = a;
    a = b;
    b = temp;
}

```

```

int main()
{
    Person people[1000000];
    int count = 0;

    ifstream fin;

    fin.open("/home/118/peoplefile50.txt");

    if (fin.fail() == true)
    {
        cout << "Can't open file :(" << endl;
    }

```

```

    while (fin >> people[count].ssn
        >> people[count].day
        >> people[count].month
        >> people[count].year
        >> people[count].first

```

```

        >> people[count].last
        >> people[count].balance)
    {
        count = count + 1;
    }

    fin.close();

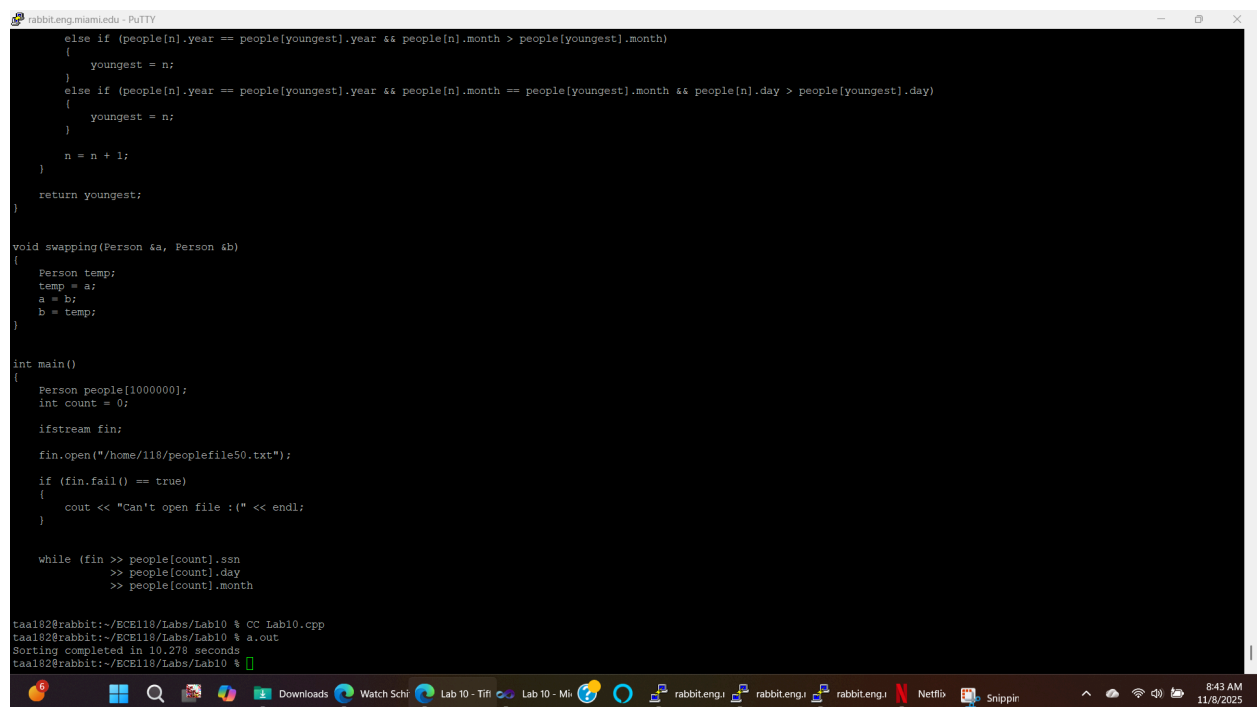
    double starting_time = get_cpu_time();

    int start = 0;
    while (start < count - 1)
    {
        int youngest = age_range(people, start, count - 1);
        swapping(people[start], people[youngest]);
        start = start + 1;
    }

    double end_time = get_cpu_time();

    cout << "Sorting completed in " << (end_time - starting_time) << " seconds" << endl;
}

```



The screenshot shows a Windows terminal window titled "rabbit.eng.miami.edu - PuTTY". The terminal displays C++ code for sorting people by age. The code includes functions for finding the youngest person in a range and swapping two people. The main function reads data from a file "peoplefile50.txt" and sorts it. The output shows the sorting completed in 10.278 seconds.

```

    else if (people[n].year == people[youngest].year && people[n].month > people[youngest].month)
    {
        youngest = n;
    }
    else if (people[n].year == people[youngest].year && people[n].month == people[youngest].month && people[n].day > people[youngest].day)
    {
        youngest = n;
    }
    n = n + 1;
}
return youngest;
}

void swapping(Person &a, Person &b)
{
    Person temp;
    temp = a;
    a = b;
    b = temp;
}

int main()
{
    Person people[1000000];
    int count = 0;

    ifstream fin;
    fin.open("/home/118/peoplefile50.txt");
    if (fin.fail() == true)
    {
        cout << "Can't open file :(" << endl;
    }

    while (fin >> people[count].ssn
           >> people[count].day
           >> people[count].month)
    {
        count++;
    }

    Sorting completed in 10.278 seconds
    taal82@rabbit:~/ECE118/Labs/Lab10 $

```

The terminal window also shows the taskbar at the bottom with various application icons and the system clock indicating 8:43 AM on 11/8/2025.

/home/118/peoplefile100.txt:

```
#include <iostream>
#include <fstream>
#include <string>
#include <time.h>
#include <sys/resource.h>
using namespace std;
```

```
double get_cpu_time()
{ struct rusage ruse;
  getrusage(RUSAGE_SELF, &ruse);
  return ruse.ru_utime.tv_sec+ruse.ru_utime.tv_usec/1000000.0 +
    ruse.ru_stime.tv_sec+ruse.ru_stime.tv_usec/1000000.0; }
```

```
struct Person
{
    string ssn;
    int day;
    int month;
    int year;
    string first;
    string last;
    double balance;
};
```

```
int age_range(Person people[], int start, int end)
{
    int youngest = start;
    int n = start + 1;

    while (n <= end)
    {
        if (people[n].year > people[youngest].year)
        {
            youngest = n;
        }
        else if (people[n].year == people[youngest].year && people[n].month >
people[youngest].month)
        {
            youngest = n;
        }
    }
}
```

```

    }
    else if (people[n].year == people[youngest].year && people[n].month ==
people[youngest].month && people[n].day > people[youngest].day)
    {
        youngest = n;
    }

    n = n + 1;
}

return youngest;
}

```

```

void swapping(Person &a, Person &b)
{
    Person temp;
    temp = a;
    a = b;
    b = temp;
}

```

```

int main()
{
    Person people[1000000];
    int count = 0;

    ifstream fin;

    fin.open("/home/118/peoplefile100.txt");

    if (fin.fail() == true)
    {
        cout << "Can't open file :(" << endl;
    }

    while (fin >> people[count].ssn
        >> people[count].day
        >> people[count].month
        >> people[count].year
        >> people[count].first

```

```

        >> people[count].last
        >> people[count].balance)
    {
        count = count + 1;
    }

    fin.close();

    double starting_time = get_cpu_time();

    int start = 0;
    while (start < count - 1)
    {
        int youngest = age_range(people, start, count - 1);
        swapping(people[start], people[youngest]);
        start = start + 1;
    }

    double end_time = get_cpu_time();

    cout << "Sorting completed in " << (end_time - starting_time) << " seconds" << endl;
}

```

```

rabbiteeng.miamiedu - PuTTY
int count = 0;
ifstream fin;
fin.open("/home/118/peoplefile100.txt");
if (fin.fail() == true)
{
    cout << "Can't open file :(" << endl;
}

while (fin >> people[count].ssn
    >> people[count].day
    >> people[count].month
    >> people[count].year
    >> people[count].first
    >> people[count].last
    >> people[count].balance)
{
    count = count + 1;
}

fin.close();

double starting_time = get_cpu_time();

int start = 0;
while (start < count - 1)
{
    int youngest = age_range(people, start, count - 1);
    swapping(people[start], people[youngest]);
    start = start + 1;
}

double end_time = get_cpu_time();

cout << "Sorting completed in " << (end_time - starting_time) << " seconds" << endl;
}

taal82@rabbitt:~/ECE118/Labs/Lab10 % CC Lab10.cpp
taal82@rabbitt:~/ECE118/Labs/Lab10 % a.out
Sorting completed in 42.3003 seconds
taal82@rabbitt:~/ECE118/Labs/Lab10 %

```